

Foundation (Jalapeno)

Q1. What is the factored form of $x^3 + y^3$?

- (A) $(x + y)(x^2 - xy + y^2)$
- (B) $(x - y)(x^2 + xy + y^2)$
- (C) $(x + y)(x^2 + xy - y^2)$
- (D) $(x - y)(x^2 - xy - y^2)$
- (E) $(x + y)(x^2 + y^2)$

Q2. What is the factored form of $8x^3 - 27y^3$?

- (A) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
- (B) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
- (C) $(2x - 3y)(4x^2 - 6xy - 9y^2)$
- (D) $(2x + 3y)(4x^2 + 6xy - 9y^2)$
- (E) $(2x - 3y)(4x^2 + 9y^2)$

Q3. What is $a^3 - b^3$ equal to?

- (A) $(a + b)(a^2 - ab + b^2)$
- (B) $(a - b)(a^2 + ab + b^2)$
- (C) $(a + b)(a^2 + ab - b^2)$
- (D) $(a - b)(a^2 - ab + b^2)$
- (E) $(a + b)(a^2 - b^2)$

Q4. Factor $64x^3 + 125y^3$

- (A) $(4x + 5y)(16x^2 - 20xy + 25y^2)$
- (B) $(4x - 5y)(16x^2 + 20xy + 25y^2)$
- (C) $(4x + 5y)(16x^2 + 20xy - 25y^2)$
- (D) $(4x - 5y)(16x^2 - 20xy - 25y^2)$
- (E) $(4x + 5y)(16x^2 + 25y^2)$

Q5. What is the formula for the sum of cubes?

- (A) $a^3 + b^3 = (a + b)(a^2 + ab + b^2)$
- (B) $a^3 + b^3 = (a - b)(a^2 - ab + b^2)$
- (C) $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- (D) $a^3 + b^3 = (a - b)(a^2 + ab - b^2)$
- (E) $a^3 + b^3 = (a + b)(a^2 + b^2)$

Q6. Factor $27x^3 - 8y^3$

- (A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
- (B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
- (C) $(3x - 2y)(9x^2 - 6xy - 4y^2)$
- (D) $(3x + 2y)(9x^2 + 6xy - 4y^2)$
- (E) $(3x - 2y)(9x^2 + 4y^2)$

Q7. What is $x^3 + 64$ equal to?

- (A) $(x + 4)(x^2 - 4x + 16)$
- (B) $(x - 4)(x^2 + 4x + 16)$
- (C) $(x + 4)(x^2 + 4x - 16)$
- (D) $(x - 4)(x^2 - 4x - 16)$
- (E) $(x + 4)(x^2 + 16)$

Q8. Factor $1000x^3 + 125y^3$

- (A) $(10x + 5y)(100x^2 - 50xy + 25y^2)$
- (B) $(10x - 5y)(100x^2 + 50xy + 25y^2)$
- (C) $(10x + 5y)(100x^2 + 50xy - 25y^2)$
- (D) $(10x - 5y)(100x^2 - 50xy - 25y^2)$
- (E) $(10x + 5y)(100x^2 + 25y^2)$

Open Ended Questions (FRQ)

Q9. Factor $8x^3 - y^3$

Q10. Factor $27x^3 + 125y^3$

Chef's Tips

- Make sure to check the answers carefully
- Encourage students to use the formulas for sum and difference of cubes
- Remind students to show their work and explain their reasoning